

CS3 – BONE CAGE FOR SPINAL IMPLANTS

PROJECT BRIEF:

To develop an interbody spinal cage to fuse the two vertebrae (spinal fusion) to treat back pain. The focus is to create a design that has an adaptive modulus, and bears an increasing load over time.

DESIGN REQUIREMENTS:

Patient Safety:

Adaptive modulus

Try to align the temporal adaptive modulus with bone formation-Degradable materials

Ability to bear an increasing load over time

Must be able to bear load until it is certain that new bone will have formed

Drive bone formation

Include design features that do this-porosity, shape, material, surface charge.

Radiolucent

Able to assess bone formation(allows x rays through so can see on scans)

KEY INFORMATION

Osteoconduction

The ability of a material or scaffold **to guide or facilitate the growth of new bone on its surface**.-needs bone
Bone cells migrate to and proliferate along the surface, forming a **scaffold for new bone formation**.

Osteointegration

Integration of the implant with surrounding bone.

- **Ensures:** **Implant** and **long-term stability**

Bone Growth Process:

Osteoblasts migrate to the implant surface → form new bone → fuse with the implant material

Osteoinduction

The ability of a tissue or material to induce bone formation by stimulating undifferentiated cells to become osteoblasts-(bone growth where there isn't surrounding bone)

Mechanics

Wolffs Law-Bone adapts to the mechanical loads it experiences:

- With regular stress, bone remodels, becoming stronger and denser-stress encourages bone growth
- With reduced stress (e.g. from a stiff implant), bone weakens and loses density

Stress shielding- **normal stress on bone** is **reduced** because an **overly stiff implant takes on** a portion of the **load**-Leads to **reduced bone density**(osteopenia).

How to Reduce Stress Shielding:

- Use more flexible or graded-stiffness materials
- Incorporate porous or lattice designs
- Match implant stiffness to surrounding bone (biomimetic design → modulus compatibility)

PATIENT SAFETY

Objective: Ensure safe load-bearing until bone is sufficiently formed.

Example q: Discuss how material and structural design choices in spinal cages ensure patient safety and effective spinal fusion
Answer structure: Material selection, Mechanical performance, Degradation risks, Patient-specific design (slot-in inserts, adaptive modulus)

BIOMIMETIC CAGE LAYOUT-LOAD DISTRIBUTION+MECHANICAL COMPATIBILITY

Material strategy: Use two complementary materials to overcome the limitations of using only one.

Outer Layer- Permanent, Hard support (Ti / Si_3N_4 / Coated PEEK)

- **Chosen material: 3D printed Ti**-high porosity-post processing to get submicron features on outside
- **Placement:** On the apophyseal ring, which bears the most load and is made of very **strong cortical bone**→ Therefore, hard material like Ti is used to **match the modulus** and provide **mechanical compatibility**.

Inner layer- Soft, resorbable support

- **Material: PART Mg alloy**, aligned with **cancellous bone** in vertebral center
 - **Placement:** Slot-in design allows **personalisation** at point of use
 - Surgeon selects fast/medium/slow resorbing insert during surgery-**matching likely biological response to the patient**
- **Adaptive Modulus:** Mg gradually degrades, transferring stress to new bone over time
- ➤ **Key Safety Need: Degradation** must be **synchronised with bone growth**→ **Coating (e.g., Chitosan)**- Slows Mg degradation to prevent premature mechanical loss (-Mg likely to degrade too quickly for new bone growth)

Bone Graft Chamber-Central slot

Central slot allows addition of bone graft material like rhBMP-2 at point of surgery

Roles:

1. **Stimulate Bone Growth** – e.g., BMPs trigger natural bone-forming pathways
2. **Provide Structural Support** – Scaffold for osteoblast attachment
3. **Promote Fusion** – Facilitates formation of solid bone bridge between vertebrae

Additional Design Features

- **Porosity & Lattice Structures:**
 - Reduce **stress shielding**
 - Improve **bone integration** and nutrient/waste flow
- **Antimicrobial Coatings:**
 - **Silver:** Prevents biofilm, shown to be **safe for bone regrowth**
 - **Copper:** May boost angiogenesis via hypoxia-mimicking signals

MATERIAL COMPARISON WHY TI AND MG?

Inner Layer: Mg Alloy vs PLA

- **Mg Alloy:**
 - Newer in some evaluation shows it upregulates/contributes to bone formation
 - **Less radiodense** than PLA → doesn't show up as much on X-rays
 - Advantage: **as Mg degrades, bone growth becomes more visible**, helping assess healing
 - **Risks:**
 - Can degrade too fast if uncoated
 - Potential cytotoxicity if too much is used
 - Requires further validation — **best used in part, not full structure**
- **PLA (Poly-lactic acid):**
 - Long clinical history
 - **More radiodense** → shows up well on scans, but can **obscure bone growth**
 - Acid buildup from degradation accelerates breakdown → feedback loop
 - **Osteoblasts don't prefer PLA** compared to Mg (cells don't like to grow on it)

- Resorption is **variable and pH-sensitive**

Outer Layer: Ti vs PEEK vs CFR-PEEK

- **Titanium (Ti):**
 - High modulus → strong support
 - **Easy to post-process (acid etch)** → **nanoscale surface** features- nanoscale features easier to do on titanium can do **post processing like acid etching, alkaline etching** etc-create **very fine pores** and grains-which **osteoblasts adhere to**
 - Best for screw fixation + **resisting mechanical failure**
- **PEEK:**
 - Radiolucent and lightweight
 - **Drawbacks:**
 - Hydrophobic → poor cell adhesion
 - **Hard to form pores** or concave shapes
 - **Cells don't like to grow on PEEK like they do on Ti.**
 - Often used for cost savings -harder to create right size and shape of pores with peek than it is with Ti-which is why market has generally moved away from peek-60% Ti 40% Peek-a lot of the people who are using it are because its slightly less costs-so in terms of manufacturing=less costs for hospital
- **Carbon Fibre-Reinforced PEEK (CFR-PEEK):**
 - Can't be 3D printed easily
 - Printing risk: fibre fragments stick out → **risk of migration in body**
 - Only safe if **fully embedded**

Mechanical Concerns with Polymers

- **Polymers like PEEK:**
 - Easier to deform over time, especially before fusion is complete
 - Poor screw holding strength
 - Degradation may reduce mechanical integrity before bone healing finishes
- **Ti Advantage:**

Stiff, durable, consistent under force- typically screws are put to stabilise cage-with polymer difficult as they are softer and weaker -undergoing that force over time-Ti much better to resist that force

- Handles long-term fixation and dynamic loading better

BONE FORMATION

Example question- **Explain how design features of spinal cages promote bone growth."**

Answer Structure: Porosity + surface shape(concave) + coating + graft + charge

- **Porous Structure:** Allows nutrients, blood vessels, and bone cells to move through the cage.(Especially needed for when material degrade-it needs to move through and cant build up)

Concave Pores: Concave Pores (Biomimetic Cues):

- Mimic osteoclast resorption pits.
- **Osteoblasts prefer curved, concave areas** — *despite Wolff's Law suggesting cell migration toward load, osteoblasts are observed to **attach and grow in unloaded concave pits.***
- Bone remodels frequently: osteoclasts dissolve bone → create concavities → osteoblasts fill in.
- **Concave pores** are essentially **biomimetic repair sites.**

Submicron/Nanoscale Features: Increase surface area → Enhance **protein adsorption** and **attracts cells** → Promote **osteinduction**

Evidence:

- **Calcium phosphate study:**
No bone growth on flat sites. However, when only the **surface features** were modified—adding **submicron pores** and creating **smart concavities** (small, curved holes)—bone formation occurred in **muscle pouches** (non-bony sites).
- **Picard et al. (2023):**
Used **titanium tubes** with concave surfaces and found that **bone formed without any surrounding bone** or chemical additives=shape driven osteoinduction
- **Surface shape alone** — especially **concave geometry** — was enough to trigger bone formation

Surface Coatings:

- **HA on Ti/PEEK** → promotes bone on-growth (not used just mentioned)
- **Nanoscale Ti on PEEK enhances bioactivity.** (easier to do with Ti)-get the benefit of cell integration onto titanium without the disadvantages of it
- **Permanent surface charge:** Attracts and binds endogenous growth factors to the surface

Biological Additions:

- **rhBMP-2** in central chamber encourages new bone formation (osteoinductive)

Piezoelectric Materials (e.g., PVDF): Create small currents during movement — shown to upregulate bone formation—this material can be used as a coat—when patients walking that deformation creates small current—attracts osteoblasts

Use of Antimicrobial Coatings: Silver & Copper (will it impact bone formation?)

Silver:

- Effective at **preventing biofilm formation** (antibacterial)
- Concern: COULD IT NEGATIVELY IMPACT OSTEOLAST ACTIVITY?
- **Evidence suggests silver is safe** — does **not inhibit bone regrowth** when used at appropriate concentrations
- Already used in several approved devices (e.g., dental implants, trauma plates)

Copper:

- Less commonly used but being explored
- May stimulate **angiogenesis** by mimicking hypoxic signals
- ongoing research efforts into using materials that release copper ions to enhance bone regeneration and fight infection
- Unclear how copper directly affects **osteoblast activity** — needs more research
- **Speculative benefit** in promoting vascularised bone healing

RADIOLUCENCY

- All polymer-based materials might offer advantages over metals or other radiopaque materials
- Better still to use adaptive radiolucent – a design that is initially radiopaque, but the resorbing elements would allow increase visualisation of new bone formation over time

Manufacture:

For hard TI layer:

Typically selective laser melting is the preferred choice, in SLM the metal powder is fully melted, however in DMLS it's partially melted.

Direct Metal Laser Sintering (DMLS):

1. DESIGN PHASE

- A **3D CAD model** of the spinal cage is created, often incorporating:
 - **Patient-specific geometry**
 - **Porous lattice structures** to mimic cancellous bone
 - Structural features for mechanical stability and bone ingrowth
- The model is digitally sliced into layers (~20–50 microns thick).

2. MATERIAL PREPARATION

- Use **medical-grade titanium powder**, typically **Ti-6Al-4V** alloy.
- The powder is highly pure, with spherical particles for uniform layering and melting.

3. DMLS/SLM PROCESS

- A **high-powered laser** in a sealed chamber selectively fuses titanium powder layer by layer.
- Each layer is:
 - Spread by a recoating blade
 - Fused where needed using the laser (melting the powder)
- The chamber is filled with inert gas (e.g., argon) to prevent oxidation.

4. BUILDING THE STRUCTURE

- The entire cage is built **layer-by-layer**, forming complex geometries such as:
 - Open-cell porous scaffolds
 - Internal support ribs
 - Anatomically matched shapes
- Wall thicknesses as thin as **0.3–0.4 mm** are achievable.

5. POST-PROCESSING

- **Heat treatment:** To reduce internal stresses
- **Support removal:** Detach build supports
- **Surface treatment to add nanoscale/submicron features:**
 - **Acid etching** (e.g., HF/HCl): creates nano-roughness
 - **Anodization:** can produce titania nanotubes
 - **Grit blasting + thermal oxidation:** improves osseointegration
- **Sterilization:** Autoclave or gamma irradiation

Benefits:

Produces **fully dense, high-strength** implants suitable for spinal loads

Patient-specific, complex geometries easily printed in SLM

Titanium cages made via SLM are **FDA-cleared and CE-marked**

2. Magnesium Alloy Inner Insert (Biodegradable Core)

- **Machining from solid bar** (most mature and safe)
- **Porosity** may be added with space-holder methods or by adjusting process parameters

B. CHITOSAN COATING (TO CONTROL DEGRADATION)

Dip-coating process:

1. Dissolve **chitosan in acetic acid solution**
 2. Dip Mg insert into the solution
 3. Dry under controlled humidity
 4. Possibly **crosslink with genipin or glutaraldehyde** to improve stability
- This coating **slows down Mg degradation** and ensures safe mechanical support until new bone grows in

3. FINAL ASSEMBLY – SLOT-IN DESIGN

- **Slot integration** is designed into the titanium cage during CAD modeling.
- Inner Mg insert is **custom-fit** and can be:
 - **Snapped or pressed in**
 - Provided in **multiple degradation rate options** (fast/med/slow)

Selective laser sintering (SLS) is commonly used for polymers, it wouldn't be compatible to manufacture the high porosity Ti or the mg alloy

While SLS can create porous plastic parts, it lacks the mechanical strength needed for load-bearing implants like spinal cages

Cant achieve high structural density like slm can.